

Ratios & Unit Rate

Use the cue, then fade the cue. Look at the cue card for 60 seconds. Then cover it before answering Round 1. Check the card only after you commit to an answer.

CUE FOCUS A ratio compares two quantities and can be written 3 to 2, 3:2, or $\frac{3}{2}$. A unit rate divides to find the amount "per one." Example: \$6 for 3 apples $\rightarrow 6 \div 3 = \$2$ per apple.

Round 1 — Retrieve It

1. What does a ratio compare?

2. Write the ratio "three to two" in three different ways.

3. What does a unit rate tell you?

4. To find a unit rate, what operation do you use and what is the "per one" amount?

Round 2 — Sketch It

Sketch a simple picture that shows \$6 shared across 3 apples. Label the total price, the number of apples, and circle the price for one apple.

Ratios & Unit Rate — Apply It

Keep the cue card covered. Solve first. Then use the cue to check, revise, and explain what you changed.

Round 3 — Apply the Relationship

1. A recipe uses 4 cups of flour to 2 cups of sugar. Write the ratio of flour to sugar in simplest form.

2. A car goes 120 miles on 4 gallons of gas. What is the unit rate in miles per gallon?

3. A 12-ounce drink costs \$3. What is the unit rate in dollars per ounce?

Round 4 — Reason from Evidence

Store A sells 5 pens for \$2.50. Store B sells 8 pens for \$3.60. Find the unit price (cost per pen) at each store, then say which store is the better deal and how you know.

Ratios & Unit Rate — ACE It

Use the cue card only if you need it. The goal is to show what you can explain, connect, and use on your own.

A

Articulate

Explain the difference between a ratio and a unit rate. Use one example that is not printed on the cue card.

C

Connect

Connect unit rate to shopping. Why does knowing the price "per one" help you compare two different-sized packages?

E

Extend

Create your own problem that asks for a unit rate. Give the numbers and show the work by dividing to get the "per one" amount.

Confidence check. Circle one after you finish: I still need the cue / I need it once in a while / I can explain this without it.

Teacher Key — Ratios & Unit Rate

Remove this page before student copying. Answers below give expected ideas, not required wording.

Retrieve It

1. Two quantities (how much of one compared to another).
2. 3 to 2, 3:2, and $\frac{3}{2}$.
3. The amount "per one" — how much of one quantity for a single unit of the other.
4. Divide; the "per one" amount is the quotient (the first quantity $\div$ the second).

Sketch It — look for

\$6 total split evenly across 3 apples, showing \$2 under each apple; the price for one apple (\$2) is circled.

Apply It

1. $4 : 2 = 2 : 1$ (flour to sugar).
2. $120 \div 4 = 30$ miles per gallon.
3. $3 \div 12 = \$0.25$ per ounce.

Reason from Evidence — look for

Store A: $2.50 \div 5 = \$0.50$ per pen. Store B: $3.60 \div 8 = \$0.45$ per pen. **Store B** is the better deal because each pen costs less.

ACE — Extend look-for

Many answers possible. Check that the problem asks for a unit rate and the work divides to get the "per one" amount.

Suggested use. Run Page 1 with the cue card available, Page 2 with the cue mostly covered, and Page 3 as independent reflection. Let students uncover the cue to self-correct in a different color or with a revision mark.